

gpu-scheduling

GPU-Aware Scheduling

Added 2026-04-17 as part of the OpenClaw Ultimate (OCU) foundation wave.

What's new

- `remote.HostResources.GPU []GPUDevice` — per-host inventory populated by `ProbeGPU` or from env labels.
- `scheduler.ContainerRequirements.GPU *GPURequirement` — optional; `nil` preserves all prior behaviour.
- `scheduler.StrategyGPUAffinity` — new placement strategy that only considers GPU-bearing hosts.
- `health.GPUHealthCheck` — VRAM-floor probe.
- `remote.ProbeGPU` — read-only SSH probe (`nvidia-smi` + `rocm-smi` + `docker nvidia runtime`); no `sudo`.

Thinker.local example

```
# .env
CONTAINERS_REMOTE_ENABLED=true
CONTAINERS_REMOTE_HOST_1_NAME=thinker
CONTAINERS_REMOTE_HOST_1_ADDRESS=thinker.local
CONTAINERS_REMOTE_HOST_1_USER=milosvasic
CONTAINERS_REMOTE_HOST_1_LABELS=gpu=true,gpu_vendor=nvidia,gpu_model=rtx3060,cuda=
CONTAINERS_REMOTE_HOST_1_GPU_AUTOPROBE=true

req := scheduler.ContainerRequirements{
    Name: "cuda-opencv",
    Image: "ghcr.io/vasic-digital/ocu-cuda-sidecar:latest",
    GPU: &scheduler.GPURequirement{
        Count: 1,
        MinVRAMMB: 2048,
        Vendor: "nvidia",
        MinCompute: "8.0",
        Capabilities: []string{"cuda", "nvenc"},
    },
}
```

```

}

sched := scheduler.NewScheduler(hm, logger,
    scheduler.WithStrategy(scheduler.StrategyGPUAffinity))
dist := distribution.NewDistributor(
    distribution.WithScheduler(sched),
    distribution.WithExecutor(sshExec),
    distribution.WithHostManager(hm),
    distribution.WithLocalRuntime(localRT),
)
summary, err := dist.Distribute(ctx,
    []scheduler.ContainerRequirements{req})

```

Backward compatibility

- `HostResources.GPU == nil` behaves identically to before.
- `ContainerRequirements.GPU == nil` means "no GPU needed"; existing strategies (`resource_aware`, `round_robin`, `affinity`, `spread`, `bin_pack`) score such requirements exactly as today.
- `Options.GPUWeight == 0` (default) means GPU score contributes 0 to total — i.e. the new code is inert until callers opt in.